

Configuring User Account Security with login.defs

Scenario

You are a system administrator for CyberShield Inc., a company that handles sensitive client data. To comply with security policies, you are Exercised with strengthening user account security by configuring parameters in the login.defs

Your goal is to ensure that all user accounts adhere to these policies for better security management.

Step 1 : Explore login.defs

1. Check the existing password, UID, and GID settings

```
grep -E  
'^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
```

2. Edit the login.defs configuration

```
sudo nano /etc/login.defs
```

Configured:

FROM:

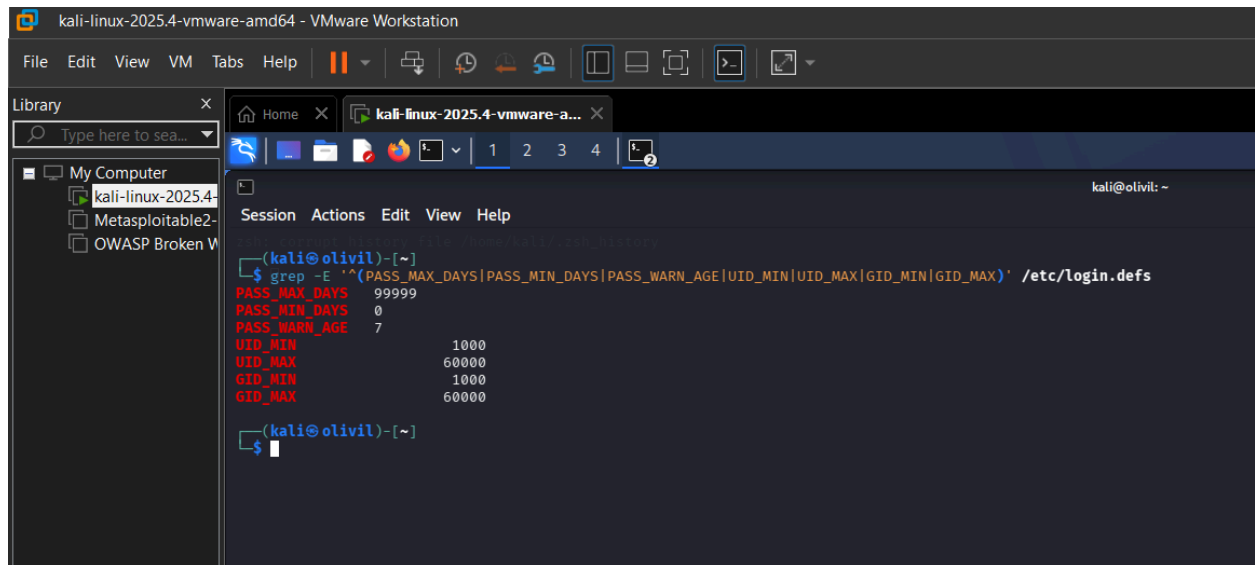
```
PASS_MAX_DAYS 99999  
PASS_MIN_DAYS 0  
PASS_WARN_AGE 7  
UID_MIN 1000  
UID_MAX 60000  
GID_MIN 1000  
GID_MAX 60000
```

TO:

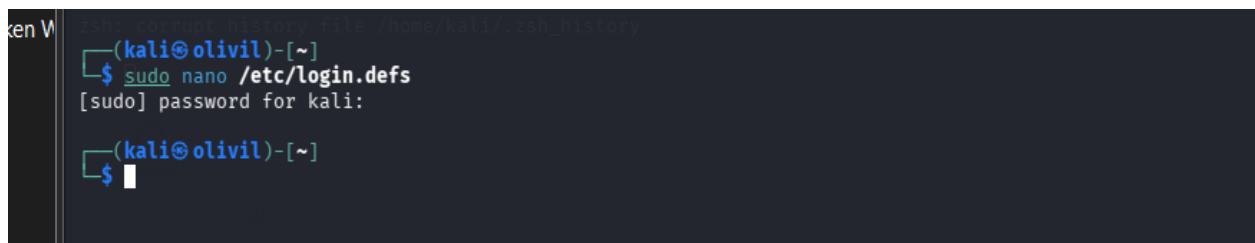
```
PASS_MAX_DAYS 90  
PASS_MIN_DAYS 7  
PASS_WARN_AGE 14  
UID_MIN 1000  
UID_MAX 60000  
GID_MIN 1000  
GID_MAX 60000
```

CTRL + O = ENTER

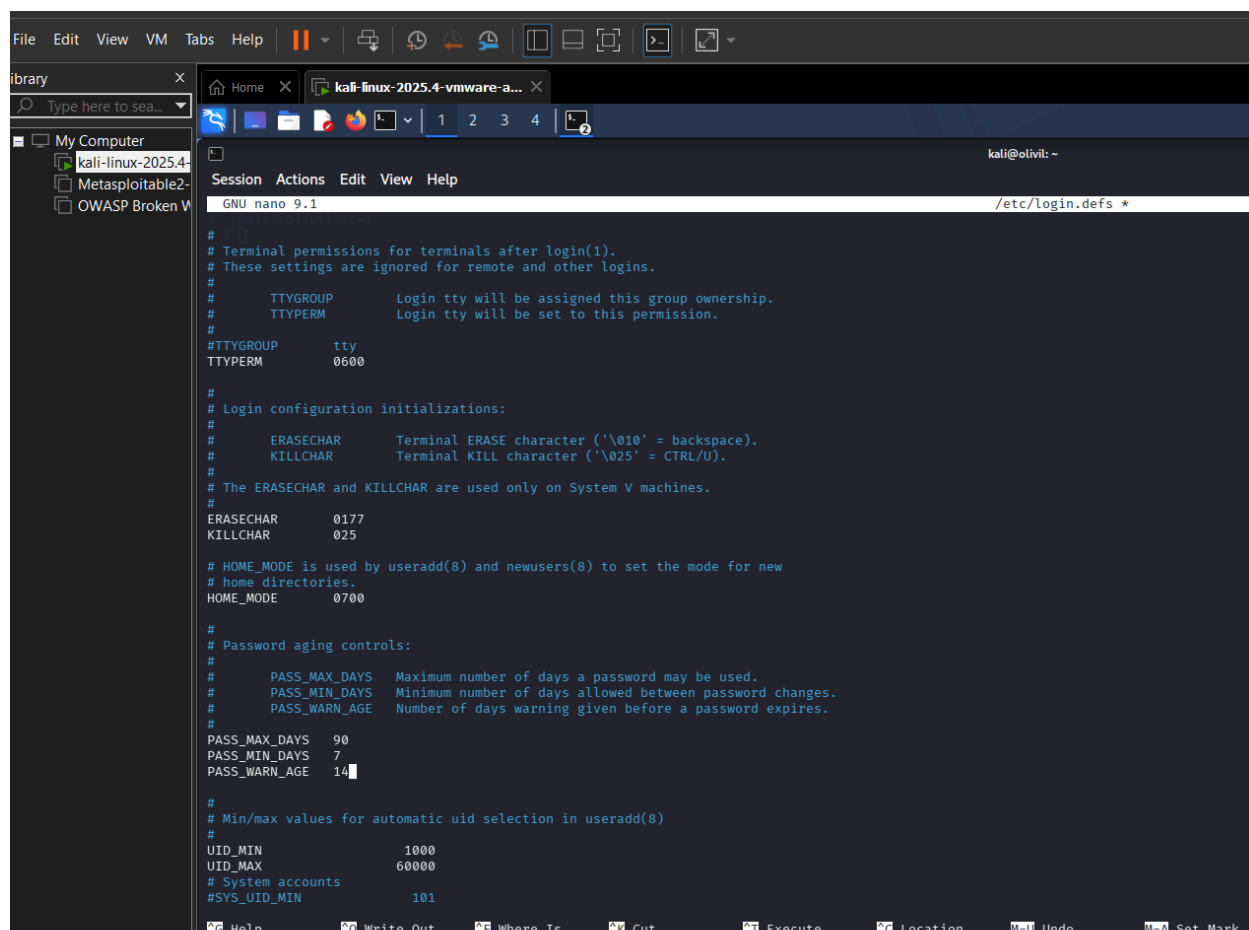
CTRL + X = EXIT FILE



```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
kali-linux-2025.4
Metasploitable2
OWASP Broken W
kali@olivil: ~
Session Actions Edit View Help
(kali@olivil)-[~]
$ grep -E '^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
PASS_MAX_DAYS 99999
PASS_MIN_DAYS 0
PASS_WARN_AGE 7
UID_MIN 1000
UID_MAX 60000
GID_MIN 1000
GID_MAX 60000
(kali@olivil)-[~]
$
```



```
ken W
(kali@olivil)-[~]
$ sudo nano /etc/login.defs
[sudo] password for kali:
(kali@olivil)-[~]
$
```



```
File Edit View VM Tabs Help
library
Type here to sea...
My Computer
kali-linux-2025.4
Metasploitable2-
OWASP Broken V
kali@olivit: ~
Session Actions Edit View Help
GNU nano 9.1 /etc/login.defs
#
# Terminal permissions for terminals after login(1).
# These settings are ignored for remote and other logins.
#
#       TTYGROUP      Login tty will be assigned this group ownership.
#       TTYPERM       Login tty will be set to this permission.
#
#TTYGROUP      tty
#TTYPERM       0600
#
# Login configuration initializations:
#
#       ERASECHAR      Terminal ERASE character ('\010' = backspace).
#       KILLCHAR       Terminal KILL character ('\025' = CTRL/U).
#
# The ERASECHAR and KILLCHAR are used only on System V machines.
#
#ERASECHAR     0177
#KILLCHAR      025
#
# HOME_MODE is used by useradd(8) and newusers(8) to set the mode for new
# home directories.
#HOME_MODE     0700
#
# Password aging controls:
#
#       PASS_MAX_DAYS  Maximum number of days a password may be used.
#       PASS_MIN_DAYS  Minimum number of days allowed between password changes.
#       PASS_WARN_AGE  Number of days warning given before a password expires.
#
#PASS_MAX_DAYS 90
#PASS_MIN_DAYS 7
#PASS_WARN_AGE 14
#
# Min/max values for automatic uid selection in useradd(8)
#
#UID_MIN       1000
#UID_MAX       60000
# System accounts
#SYS_UID_MIN   101
```

First reviewed the existing login.defs settings to establish the current account-security configuration. Then opened the file and updated the password policy to require a maximum password age of 90 days, a minimum age of 7 days, and a 14-day warning period. The existing UID and GID ranges already matched the required security policy, so they were left unchanged.

Step 2: Tested User and Applied Password Policy

Commands used:

1. Verify the updated login.defs settings

```
grep -E  
'^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
```

2. Attempt to create the test user

```
sudo useradd testuser (which user already exist from previous labs)
```

3. Verified the existing testuser account

```
id testuser
```

4. Checked the user's original password-aging settings

```
sudo chage -l testuser
```

5. Applied the required password policy

```
sudo chage -m 7 -M 90 -W 14 testuser
```

6. Verified the updated password-aging settings

```
sudo chage -l testuser
```

```
(kali@olivil)-[~]
$ grep -E '^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
PASS_MAX_DAYS    90
PASS_MIN_DAYS     7
PASS_WARN_AGE    14
UID_MIN           1000
UID_MAX          60000
GID_MIN           1000
GID_MAX          60000

(kali@olivil)-[~]
$ sudo useradd testuser
useradd: user 'testuser' already exists

(kali@olivil)-[~]
$ id testuser
uid=1001(testuser) gid=1001(testuser) groups=1001(testuser),100(users)

(kali@olivil)-[~]
$ sudo chage -l testuser
Last password change                : May 11, 2026
Password expires                    : Feb 23, 2300
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7

(kali@olivil)-[~]
$ sudo chage -m 7 -M 90 -W 14 testuser

(kali@olivil)-[~]
$ sudo chage -l testuser
Last password change                : May 11, 2026
Password expires                    : Aug 09, 2026
Password inactive                   : never
Account expires                     : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14

(kali@olivil)-[~]
$
```

verified that the required login.defs settings were correctly configured. The testuser account already existed, so instead of creating a duplicate account, I checked its existing settings and applied the required password policy using **chage**. The final verification confirmed a 7-day minimum password age, 90-day maximum password age, and 14-day warning period.

Step 3: Set and Test the testuser Password

Commands used:

1. Set a password for testuser

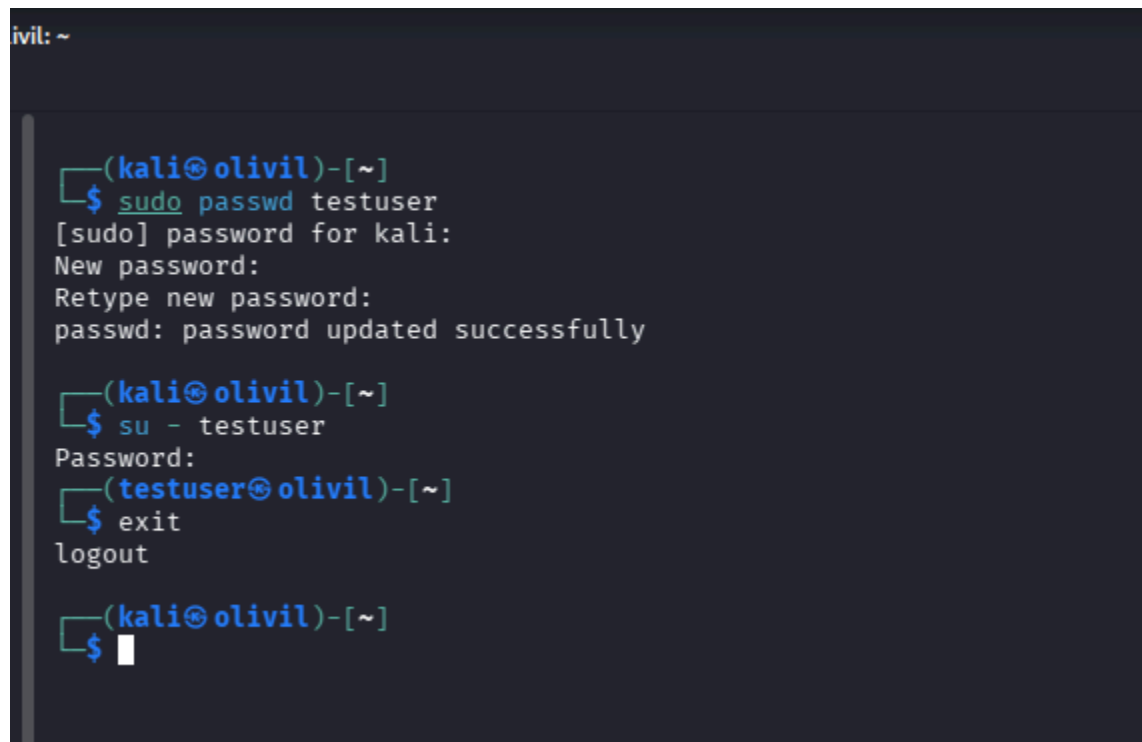
`sudo passwd testuser`

2. Switch to the testuser account

`su - testuser`

3. Exit the testuser session

`exit`



```
ivilt: ~  
  
(kali@olivil)-[~]  
$ sudo passwd testuser  
[sudo] password for kali:  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(kali@olivil)-[~]  
$ su - testuser  
Password:  
(testuser@olivil)-[~]  
$ exit  
logout  
  
(kali@olivil)-[~]  
$
```

successfully set a password for the testuser account and verified that the account could be accessed by switching to it with `su - testuser`. After confirming successful login, I exited the test account and returned to the normal kali user session.
